MILITARY SPECIFICATION

TRAINER, BASIC, ELECTRONICS, STUDENT

(PHASE II)

This specification has been approved by the Office of Naval Research (U. S. Naval Training Device Center), Department of the Navy.

1. SCOPE

1.1 This specification covers the design development and construction of additional subchassis for the Basic Electronics Student Trainer (BEST) Device 6B19 (NAVPERS 17540A).

1.1.1 This item has been designated by the Office of Naval Research (U. S. Naval Training Device Center), as Device 6B19 (NAVPERS 17540A) Phase II for Navy procurement (see 6.2). Device 6B19 (NAVPERS 17540A) Phase I procurement is covered in MIL-T-82330.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard

Military

MIL-R-11	Resistors, Fixed, Composition, (Insulated), General Specification For
MIL-C-25	Capacitors, Fixed, Paper-Dielectric, Direct Current (Hermetically Sealed in Metallic Cases), General Specification for

FSC 6910

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MIL-P-116	Preservation; Methods of
MIL-S-3950	Switches, Toggle
MIL-A-8625	Anodic-Coatings for Aluminum and Aluminum Alloys
MIL-E-15090	Enamel, Equipment, Light Gray (Formula No. 111)
MIL-E-16400	Electronic Equipment, Naval Ship and Shore; General Specification For
MIL-T-23991	Training Devices, Military; General Specification For
MIL-T-82330	Trainer, Basic Electronics, Student (Phase I)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-440	Soldering Techniques for Standard Type Solder Terminals
MIL-STD-681	Identification Coding and Application Hookup and Lead Wire
MIL-STD-749	Guidelines for Submission of Data on Non-Standard Electronic Parts

DRAWINGS

U. S. Naval Training Device Center

335-100-6B19	Schematics, Wiring Diagrams and Parts Lists for Device 6B19

PUBLICATIONS

U. S. Naval Training Device Center

335-101-6B19	Test Procedures for Device 6B19

(Copies of specifications,standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be selected in accordance with MIL-T-23991.

3.2 Design.- The subchassis shall consist of various electronic circuits as specified in 3.5 through and including 3.5.10.3. The subchassis shall be powered through a modified cannon plug (see 3.5.19) which shall be affixed to the subchassis. The affixed and modified cannon plug wiring shall conform to its mating cannon plug on the existing basic power unit and the power adapter

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plug units of MIL-T-82330. The power supply to the adapter plug units shall be furnished by the master power unit of MIL-T-82330.

3.2.1 Reliability of operation.- Reliability of operation requirements of MIL-T-23991 shall apply.

3.2.2 Design data.- Prior to the actual purchase of hardware and fabrication of Device 6B19 (NAVPERS 17540A) Phase II, design data shall be prepared in the form of a report for review and approval of the contracting officer and shall consist of engineering, electrical and mechanical drawings, schematics, sketches and descriptive data.

3.2.3 Interchangeability.- All parts, subassemblies, and assemblies having the same part number shall be directly and completely interchangeable with each other, in respect to installation and performance, regardless of sources from which obtained, without alteration and without selection for fit or performance.

3.2.4 Safety.- With the exception of the interlocks requirement, components shall be installed to comply with the safety requirements of MIL-E-16400 and as specified in 3.2.4.1 through and including 3.2.4.3

3.2.4.1 Klystron tube cover.- The klystron tube external surface shall be covered with a plastic shield.

3.2.4.2 Klystron output.- The klystron transmitter output shall be designed with minimum power output to prevent radiation hazards to operating personnel.

3.2.4.3 Klystron warning signs.- The following warning signs shall be installed:

(a) A warning sign shall be installed on the plastic cover indicating that the klystron shell is energized.

(b) A warning sign shall be installed on the klystron chassis indicating that the klystron filament shall have a one minute warmup before the application of repeller and shell voltages.

3.3 Construction.- Construction shall be in accordance with the following:

3.3.1 Component mounting boards.- Component mounting shall be of 3/32-inch melamine.

3.3.2 Component layout.- Component layout shall comply with the techniques, general arrangement and accessability of components as shown on the drawings of 335-100-6B19.

3.3.3 Soldering.- Soldering techniques shall comply with MIL-STD-440.

3.3.4 Wiring.- Subchassis wiring shall be solid wire of a size to meet subchassis power requirements. Wiring shall comply with the slack, insulation protection, connections and wire terminations requirements of MIL-E-16400. Stranded wire shall be used where flexing will cause breakdown of the insulation. Multiple wires shall run close to the chassis, observing standard feedback prevention, and laced to comply with the internal cabling requirements of MIL-E-16400. Circuit ground connections shall be from a wire, to

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a solder lug, to the chassis. No solder connections to the chassis shall be made.

3.3.4.1 Wire color coding.- Differentiation color coding of chassis wiring shall be used for identification of circuits and checkpoints to comply with MIL-STD-681.

3.4 Performance.-

3.4.1 The subchassis as listed in 3.5 shall be capable of operation, as (a) a single subchassis (b) a combination of subchassis, or (c) as a specific system in combination with the subchassis of MIL-T-82330, as outlined in a course curriculum. Each plug-in subchassis shall be capable of being unplugged from its power receptable plug unit, rotated 180°, and replugged to assume a back-to-front position for the purpose of gaining access to the subchassis' rear connections.

3.4.2 Controls and components.- All controls and movable components shall operate smoothly without binding. The controls and components, when set to a selected position, shall not reset to a different position until moved manually.

3.5 Details of subchassis.- The subchassis shall consist of the following, the details of which shall be as specified in 3.5.1 through and including 3.5.10.3:

Description	NAVTRADEVCEN No.	NAVPERS No.
(a) Vacuum Tube Analyzer	6B19/9	17540A-9
(b) Transistor Analyzer	6B19/10	-10
(c) Transistor Receiver	6B19/11	-11
(d) Frequency Modulation (F.M.) Detector	6B19/23	-23
(e) Magnetic Amplifier	6B19/24	-24
(f) Delay Line	6B19/12	-12
(g) Klystron	6B19/14	-14
(h) I.F. Amplifier (VIDEO)	6B19/15	-15
(i) Square Wave Generator	6B19/19	-19
(j) Counter	6B19/25	-25

3.5.1 Vacuum tube analyzer.- This unit shall provide the means to demonstrate static characteristic measurements of typical pentode and triode vacuum tubes. The vacuum tube analyzer subchassis shall consist of the following main components:

3.5.1.1 Vacuum tubes.-

(a) One 6AU6 pentode tube
(b) One 6J6 triode tube.

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3.5.1.2 Meters.-

(a) A 1 3/4-inch edgewise type milliammeter
(b) A 1 3/4-inch edgewise type plate voltmeter
(c) A 1 3/4-inch edgewise type grid voltmeter
(d) A 1 3/4-inch edgewise type milliammeter installed in the cathode of the 6AU6 pentode tube.

3.5.1.3 Switches and controls.-

(a) A switch to connect power and the meters of 3.5.1.2 between the elements of the triode and pentode
(b) A wire wound potentiometer to control the filament voltage
(c) A 1 3/4-line edgewise type grid voltmeter.
(d) A composition potentiometer to control the bias voltage.

3.5.2 Transistor analyzer.- This unit shall provide the means to demonstrate input-output characteristics, forward and reverse bias and direct current alpha and beta measurements of low power transistors. The transistors used shall be comparable to those used in the transistor receiver of 3.5.3. The transistor analyzer subchassis shall consist of the following main components:

3.5.2.1 Transistors.-

(a) One NPN transistor
(b) One PNP transistor.

3.5.2.2 Meters.-

(a) An emitter milliammeter
(b) A center zero base microammeter
(c) A collector milliammeter
(d) A center zero voltmeter with appropriate meter shunt, shall be provided, to retain compatible deflection when switching between collector and emitter.

3.5.2.3 Switches and controls.-

(a) A switch to connect power and the meters of 3.5.2.2 to the NPN or the PNP transistor.
(b) A switch to connect the voltmeter of 3.5.2.2(d) to the collector or the emitter of the transistors.
(c) A switch to change the circuit configuration to either common emitter, common base, or common collector
(d) A composition potentiometer switched between the collectors of the NPN and PNP transistors for varying the potential

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(e) A switch to permit intercharge of the potentiometer of 3.5.2.3(d).
(f) A composition potentiometer to control voltage of emitters of both the NPN and PNP transistors.
(g) An internal Zener diode regulation circuit to insure constant voltage to the transistors.

3.5.3 Transistor receiver.- This unit shall provide the means for the student to understand the operation and apply the techniques used to trouble-shoot and calibrate circuits using solid-state components. This unit shall be a superheterodyne, covering the standard broadcast frequencies and shall include the following:

(a) One radio frequency (RF) stage
(b) One mixer stage
(c) Two oscillators (Hartley and Colpitts types)
(d) Two intermediate frequency stages (IF)
(e) A diode detector with simple and delayed automatic volume control (AVC)
(f) A push-pull audio output stage comparable in circuit configuration to the Audio Amplifier 6B19/8 (NAVPERS 17540A-8). A suitable speaker shall be provided.

3.5.3.1 Switches and controls.- The unit shall be developed with:

(a) A power ON-OFF switch ganged with a volume control
(b) A variable tone control
(c) A main variable capacitor provided with a calibrated dial having oversized numerals at 600 kilocycles (kc) and 1400 kc.
(d) A switch to select either the Hartley or Colpitts Oscillator
(e) A switch to select either simple or controlled delayed AVC
(f) A switch in each emitter to open circuit and connect in a milliammeter.

3.5.3.1.1 Adjustments.- All adjustments for the antenna, R. F., oscillators and I.F. circuits shall be standard screwdriver adjustable type and available at the front panel.

3.5.3.2 Test points.- Front panel color coded test jacks shall be provided in the following circuit locations.

(a) From each side of the power ON switch (to provide connection of a test milliammeter).
(b) To emitter, base and collector points of each transistor
(c) To AVC bus
(d) To supply voltage (22V)
(e) To ground point
(f) To each side of the speaker voice coil (for monitoring calibration levels)

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(g) To the antenna input, through a gimmick coupling coil, (to provide connection to a signal generator)

(h) To each side of the emitter switching points, to connect to a milliammeter.

3.5.4 F.M. detector.- This unit shall provide the means for the student to understand the operation and apply the techniques used to trouble shoot and calibrate the various types of F.M. detectors. The unit shall be developed with circuit configurations, each circuit having a slug-tuned input of 30 megacycles (MC), as follows:

(a) A pentode high frequency (HF) amplifier whose output shall be capable of being switched to either the inputs of the limiter, the discriminator, or the ratio detector.

(b) A pentode limiter circuit whose input shall be capable of being switched (see 3.5.4.1) to either the output of the HF amplifier or to an OFF position. The output of the pentode limiter circuit shall be fed through a ganged switching arrangement to either the input of the discriminator circuit or to an OFF position.

(c) A balanced/unbalanced ratio detector circuit whose input shall be capable of being switched to either the output of the HF amplifier or to an OFF position.

(d) A discriminator detector circuit whose input shall be capable of being switched to either the output of the HF amplifier, to the output of the pentode limiter circuit or to an OFF position.

3.5.4.1 Switches.- The FM detector subchassis shall be equipped with the following controlling switches on the front panel.

(a) A multiwafer and multipole switch connecting the circuits described in 3.5.4 (a), (b), (c) and (d). The switch shall also apply power to the selected circuits.

(b) A switch shall be provided in the circuitry to permit disconnecting of the limiter action capacitor to permit S curve tests.

(c) A switch to select the audio output of the discriminator or the ratio detector, at either the balanced or unbalanced points in the circuit.

3.5.4.2 Test points.- Front panel test jacks shall be provided in the following circuit locations: (a) At the output of the discriminator. (b) At the output of the limiter (c) At the output of the ratio detector at its balanced and unbalanced points.

3.5.4.3 BNC connecters.- Front panel BNC connecters shall be provided in the following circuit locations:

(a) The input of the HF amplifier
(b) The audio output of the discriminator
(c) The audio output of the ratio detectors.

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3.5.4.4 Additional features.- Additional features shall be incorporated in the FM detector subchassis as follows:

(a) A shielded connection from the input of the HF amplifier to pins no. 2 and 15 of the cannon connecter plug of the subchassis.
(b) A shielded connection between the selected output of switch 3.5.4.1 (c) to pins no. 3 and 14 of the cannon connecter plug of the subchassis.

3.5.5 Magnetic amplifier.- The magnetic amplifier subchassis shall provide the means for the student to understand the various types, the operation, and practical application of saturable reactors and magnetic amplifiers. The magnetic amplifier subchassis shall be developed with the following circuit capabilities:

(a) Half-wave saturable reactor
(b) Full-wave saturable reactor
(c) Half-wave magnetic amplifier
(d) Full-wave magnetic amplifier

3.5.5.1 Circuit features.- The magnetic amplifier subchassis shall be provided with necessary test points BNC connecters, front panel controls and switches to permit changes in control currents, bias and feedback to enable the demonstration of the following circuit parameters:

(a) Series aiding or opposing series windings
(b) Internal or external feedback
(c) Regenerative or degenerative feedback.

3.5.5.2 Applications.- The magnetic amplifier subchassis shall be capable of being utilized as a:

(a) Voltage regulator (DC)
(b) Voltage regulator (AC)
(c) Audio amplifier
(d) Series control unit
(e) Light dimmer
(f) Motor speed control
(g) Switching, memory or logic circuit.

3.5.6 Delay line.- The delay line subchassis shall provide the means for the student to understand the operation and perform measurements of lumped parameter or electromagnetic delay lines and shall be capable of demonstrating the following:

(a) Open-ended delay line
(b) Shorted delay line.

3.5.6.1 Additional features.- The following features shall be provided;

(a) The delay line shall be designed to produce discernable delays

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in five variable steps on an oscilloscope with a time base frequency of 5 cycles per second (CPS) to 1 megacycle (MC).

(b) The delay line shall be capable of being used as a pulse forming network.

3.5.6.1.1 Front panel switches and connections.-

(a) A rotary switch shall be provided and each position labeled with the value of L and C and the delay time in microseconds.

(b) A toggle switch shall be used for shorting and opening the delay line.

(c) An input and output BNC connector shall be provided on the front panel

3.5.7 Klystron.- The klystron subchassis shall provide the means for the student to understand the operation and safely perform the measurements of micro-wave transmitters and receiver conversion circuits. The subchassis shall incorporate the following circuitry:

(a) An X-band reflex klystron with a resonant cavity that is capable of operating as a local oscillator and as a transmitting klystron. A training technique shall require one klystron unit to transmit intelligence to another klystron unit.

(b) As a local oscillator klystron the unit shall include a balanced silicon crystal mixer within an X-band antenna horn. When the transmitted radar signal is received the unit shall develop 30 mc I.F. The unit shall provide a metering circuit for the crystal mixer.

(c) As a transmitter, the Klystron repeller shall be capable of accepting an externally applied pulse from a master timing chain, an audio or a pulse circuit thereby permitting it to transmit intelligence through the X-band antenna horn.

(d) As a local oscillator or transmitter, the klystron unit shall operate from the available power supplies NAVPERS 17540A-1 and 28.

(e) The klystron coarse frequency control shall be by resonant cavity screwdriver adjustment. The fine frequency control shall be by potentiometer reflector (repeller) voltage adjustment.

3.5.7.1 Test points.- Front panel test jacks and BNC connecters shall be provided to:

(a) Connect the external modulating source

(b) Connect the test oscilloscope to view voltage modes and klystron bandwidth.

(c) Measure voltages at the reflector (repeller), shell, and cathode of the klystron.

(d) Connect the I.F. output.

3.5.8 Video I.F. amplifier.- The video I.F. amplifier subchassis shall provide the means for the student to understand the operation and apply the techniques used to trouble shoot and calibrate video I.F. amplifiers. The subchassis shall incorporate the following circuitry:

(a) A three stage cascade amplifier demonstrating the characteristics

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of cascade amplifiers.

(b) A one stage cascode amplifier demonstrating the tuning and gain characteristics of a grounded grid cascode amplifier

(c) The amplifiers of (a) and (b) shall have a bandwidth of 2 MC at a frequency of 30 MC.

3.5.8.1 Additional circuit requirements.- The circuits of 3.5.8 shall have the following features:

(a) Untuned I.F.
(b) Single-tuned I.F.
(c) Double-tuned I.F.
(d) Stagger-tuned I.F.
(e) Variable gain
(f) Variable bandwidth (with swamping resistors)
(g) Input to be compatible to klystron unit of 3.5.7.
(h) Output to be compatible to Detector Unit 6B19/7 (NAVPERS 17540A-7).

3.5.8.2 Switches and controls.- The video subchassis shall be developed with the following front panel controlled components:

(a) A switch to connect/disconnect the cascode I.F. amplifier in the cascade I.F. string.

(b) A multipole switch to select any single or combination of circuit features of 3.5.8.1

(c) A switch to select the capability of 3.5.8.1 (f)

(d) A composition control to perform 3.5.8.1 (e) function

3.5.8.3 BNC connecters.-Front panel connecters shall be provided at the following circuit points:

(a) Input to the cascade amplifier
(b) Input to the cascode I.F. amplifier
(c) Output of the cascade I.F. amplifier
(d) Each grid of the cascade I.F. amplifier.

3.5.8.4 Additional features.- A coaxial lead shall be connected from the output of the cascade amplifier to pins no. 3 and no. 14 of the cannon connecter plug of the subchassis.

3.5.9 Square wave generator.- The square wave generator subchassis shall provide the means for the student to understand the operation of overdriven amplifiers, wave shaping circuits and components. The subchassis shall be developed demonstrating the following:

(a) An overdriven vacuum tube
(b) An overdriven power transistor

3.5.9.1 Circuit capabilities.- The square wave generator subchassis circuits shall provide the following:

(a) Square wave output

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